

Calculus II

Area locked by curve

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2019

Outline

- 1 Areas Locked by Curves
- 2 Areas in Polar Coordinates

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Areas

- The area under a curve $y = F(x)$ from a to b is

$$A = \int_a^b F(x)dx$$

- Suppose the curve has parametric equations $x = f(t)$, $y = g(t)$, $\alpha \leq t \leq \beta$.
- Then use the Substitution Rule to find the area:

$$A = \int_a^b ydx = \int_{\alpha}^{\beta} g(t)f'(t)dt$$

- How do we know where to put α and β ?
- When $x = a$, t will be either α or β . When $x = b$, t will take the other value.

Example

Find the area under one arch of the cycloid

$$x = r(\theta - \sin \theta), \quad y = r(1 - \cos \theta)$$

Areas in Polar Coordinates

Suppose we have a polar curve $r = f(\theta)$, $a \leq \theta \leq b$.

Definition

We say that the figure obtained as the union of the segments connecting the origin with the points of the curve is the figure *swept* by the curve as θ varies from a to b .

Theorem

Suppose no two points on the curve lie on the same ray from the origin. Then the area swept by the curve equals $A = \int_a^b \frac{1}{2} (f(\theta))^2 d\theta$.

Area swept by a polar curve: justification

Split $[a, b]$ into N equal segments via points $a = \theta_0 \leq \theta_1 \leq \dots \leq \theta_{N-1} \leq \theta_N = b$. The length of each segment is $\Delta = \frac{b-a}{N}$. Let $r_i = f(\theta_i)$. Then each θ_i gives a point P_i with polar coordinates (r_i, θ_i) .

The area swept by the curve is approximated by sum of areas of triangles given by connecting the origin with two consecutive vertices. Consider one such triangle, say, OP_1P_2 . By Euclidean geometry, the area of $\triangle OP_1P_2$ is $\frac{|OP_1||OP_2|\sin \Delta}{2} = \frac{r_1 r_2 \sin \Delta}{2} = \frac{f(\theta_1)f(\theta_2) \sin \Delta}{2}$.

Area swept by a polar curve: justification

Split $[a, b]$ into N equal segments via points $a = \theta_0 \leq \theta_1 \leq \dots \leq \theta_{N-1} \leq \theta_N = b$. The length of each segment is $\Delta = \frac{b-a}{N}$. Let $r_i = f(\theta_i)$. Then each θ_i gives a point P_i with polar coordinates (r_i, θ_i) .

Therefore the area swept by the curve equals the limit of the sum:

$$\begin{aligned}
 A &= \lim_{\Delta \rightarrow 0} \sum_{i=0}^{N-1} \frac{f(\theta_i)f(\theta_{i+1})\sin \Delta}{2} = \lim_{\Delta \rightarrow 0} \frac{\sin \Delta}{\Delta} \sum_{i=0}^{N-1} \frac{f(\theta_i)f(\theta_{i+1})\Delta}{2} \\
 \text{(can be proved)} &= \lim_{\Delta \rightarrow 0} \frac{\sin \Delta}{\Delta} \lim_{\Delta \rightarrow 0} \sum_{i=0}^{N-1} \frac{f(\theta_i)f(\theta_{i+1})\Delta}{2} = 1 \cdot \lim_{\Delta \rightarrow 0} \sum_{i=0}^{N-1} \frac{f(\theta_i)f(\theta_i)\Delta}{2} \\
 \text{(Riemann sum)} &= \lim_{\Delta \rightarrow 0} \sum_{i=0}^{N-1} \frac{f^2(\theta_i)\Delta}{2} = \int_a^b \frac{f^2(\theta)}{2} d\theta
 \end{aligned}$$

Example

Find the area enclosed by one loop of the four-leaved rose $r = \cos 2\theta$.

The region enclosed by the right loop corresponds to points whose θ polar coordinate lies in the interval $-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}$.

$$\begin{aligned} A &= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{2} r^2 d\theta \\ &= \frac{1}{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos^2(2\theta) d\theta \\ &= \int_0^{\frac{\pi}{4}} \cos^2(2\theta) d\theta \\ &= \int_0^{\frac{\pi}{4}} \frac{1}{2} (1 + \cos(4\theta)) d\theta \\ &= \frac{1}{2} \left[\theta + \frac{1}{4} \sin(4\theta) \right]_0^{\frac{\pi}{4}} \\ &= \frac{\pi}{8} \end{aligned}$$

Example

Find all points of intersection of the polar curves $r = \frac{1}{2}$ and $r = \cos(2\theta)$.

$$\cos 2\theta = \frac{1}{2}$$

$$2\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

- This only gives four points.
- There are actually eight.
- The circle $r = \frac{1}{2}$ also has polar equation $r = -\frac{1}{2}$.
- To find all eight points, solve $\cos(2\theta) = \frac{1}{2}$ and $\cos(2\theta) = -\frac{1}{2}$.

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

The curves meet if

$$3 \sin \theta = 1 + \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left(3 \frac{\pi}{2} - 2 \cdot 0 + 2 \cdot 0\right) - \left(3 \frac{\pi}{6} - 2 \frac{\sqrt{3}}{2} + 2 \frac{\sqrt{3}}{2}\right) \\ &= \pi \end{aligned}$$